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Bachelor of Science in Information Technology

ResQNet PH: A Real-Time Flood Monitoring and Community Alert System Using Crowdsourced Data and Predictive Analytics

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BSIT 1.3

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Date:

May 17, 2026

Final Project/Exam: C++ Solution for Sustainable Development

Section 1: Introduction

1.1 ResQNet PH is a C++ console utility designed to bridge the gap between national weather advisories and barangay-level reality.

1.2 Why Does This Software Need to Exist?

Even though the Philippines has national weather agencies like PAGASA, there is still a "black hole" of information when floods hit local neighborhoods. **ResQNet PH** exists because of three main problems:

- **Missing Neighborhood Details:** National reports are too broad for a specific street. While we might know it is raining in the whole city, residents don't have the **specific data (like exact river levels)** needed to know if their own street is still safe.
- **Rumors and Fake News:** Because there is no official local app, people rely on social media for updates. This leads to **unverified rumors and confusion**, which can cause people to panic or, even worse, stay in danger for too long.
- **Acting Too Late:** Without a central system, local rescue teams usually **react only after the flood has already happened**. Decisions are made too late, which puts lives and property at much higher risk.

- **Section 2: Requirements Analysis**

2.1 Functional Requirements (FR): What the system *does* (e.g., "FR1: System must calculate carbon footprint").

FR 1: Flood Risk Assessment

The system must analyze input data such as water level and classify areas into **Low, Moderate, or High Risk**.

FR 2: Data Input Handling

The system must allow users to input flood-related data (sample. Water level, Location).

FR 3: Safe Route Identification

The system must determine safe and passable routes based on current flood conditions.

FR 4: Alert Generation

The system must display warnings or alerts when an area is classified as high risk.

FR 5: Data processing and Output

The system must process all inputs and display clear, readable results for users.

2.2 Non-Functional Requirements (NFR): Performance, Robustness (Input Validation), and Readability.

- **Performance**

The system should provide fast and real-time results with minimal delay.

- **Accuracy**

The system must produce reliable and correct flood risk classification.

- **Usability**

The interface should be simple and easy to understand even for non-technical users.

- **Robustness**

The system must handle invalid or incorrect inputs without crashing.

- **Readability and maintainability**

The code must be clean, organized and easy to update or improve.

- **Section 3: Design Specification**

3.1 **Algorithmic Flowchart:** A detailed flowchart of the system's most complex logic.

START

|

Initialize ResQNet System

(FloodMonitor monitor — displayHeader())

|

Load Existing Zone Data (Auto on Startup)

loadFromFile("database.txt") — runs before menu appears

Skips silently if file not found

|

Display Main Menu (Switch Case Driver)

[1] Input / Add Flood Zone Report

[2] Generate Predictive Risk Assessment

[3] View Safe Route Suggestions

[4] CRUD Zone Management

[5] Search Zone by ID

[6] Search Zone by Name

[7] Statistics Dashboard

[0] Save & Exit

|

Select Option (0–7)

|

[If Option 1 Selected] → logFloodReports()

|

Input Zone Information:

- Zone Name (Barangay)

- Municipality

- Water Level (cm) — validated via cin loop

- Number of Families Affected

- Evacuation Status (y/n)

- Rescue Team Assigned

|

Validate Input

|

Is data valid? (isValidLevel — 0 to 500 cm)

| — NO → [!] Invalid. Must be 0–500 cm → Return to Input

| — YES

|

Compute Risk Indicators

generateNewId() — assign unique zone ID

|

Check Critical Condition:

Is Water Level ≥ 200 OR Families ≥ 600 ?

| — YES

| |

| Set Alert Level = RED

| Set Priority = HIGH

| Create CriticalZone object

| Recommend Immediate Evacuation

|

└─ NO

|

Check Low Risk:

Is Water Level < 50 AND Families < 200?

└─ YES

| |

| Set Alert Level = GREEN

| Set Priority = LOW

| Create MonitoredZone object

| No Evacuation Needed

|

└─ NO

|

Set Alert Level = ORANGE

Set Priority = MEDIUM

Create FloodZone (base) object

Prepare Evacuation Plan

|

Check Evacuation Status

|

Is Evacuation = YES?

└─ YES → Assign Rescue Team & Mark as "In Progress"

└─ NO → Zone not yet evacuated — Monitor Continuously

|

Update RESQNET Database / Zone Record

zones[] vector — push new FloodZone* pointer

|

Save Data to File (Persistent Storage)

saveToFile() — write all zones to database.txt

|

Display Output:

- Zone Info (ID, Name, Municipality)
- Alert Level (RED / ORANGE / GREEN)
- Priority Level (HIGH / MEDIUM / LOW)
- Rescue Team & Evacuation Status
- z->printDetails() — polymorphic call

|

Return to Main Menu / Process Another Zone?

└— YES → Go back to Switch Case Driver / Input

└— NO

|

Terminate System Session

exitAndSave() — final save + goodbye message

|

END

3.2 File Structure Diagram: Explanation of the data format used in external files.

The system uses a structured file format to store and manage flood monitoring data efficiently. All zone records are saved in an external file “Zones.txt.” using a table-like structure where each row represents a single zone and each column represents a specific data field

ZoneID	ZoneName	Municipality	WaterLevel	Families	Evacuated	AlertLevel
Z001	Riverside	Manila	210	650	Yes	RED
Z002	Poblacion	Quezon	120	300	No	ORANGE
Z003	San Isidro	Bulacan	40	150	No	GREEN

ZoneID – Unique identifier for each area

ZoneName – Name of the zone or location

Municipality – City or province where the zone is located

WaterLevel – Measured water level (in centimeters)

Families – Number of affected families

Evacuated – Indicates whether evacuation has started (Yes/No)

AlertLevel – System generated risk level (GREEN, ORANGE, RED)